

Wound Care Advisor

Planning & Design Overview

Hospital in the Home — Community Nursing

Version 1.0 · June 2026 · Internal Planning Document

Purpose of this document

This document provides a lay-person's overview of how the Wound Care Advisor mobile tool was conceived, planned, and structured — from a hand-sketched idea to a deployable single-file web application.

It is intended for: clinical leads, project sponsors, IT approvers, and new team members who need to understand what the tool does and how it is organised without reading code.

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1 The Problem Being Solved

What gap does this tool fill?

Community nurses visiting patients at home — as part of Hospital in the Home (HITH) programs — must make real-time wound dressing decisions, often without immediate access to a wound specialist or printed reference charts. The Australian Government Department of Health and Aged Care (DoH) publishes a Consumable Choice Guide: a 26-pathway matrix that maps wound characteristics to appropriate dressing types. However, this matrix exists as a printed chart that is difficult to use quickly at the bedside, especially on a mobile phone.

Challenge	How the tool addresses it
Printed DoH matrix hard to navigate on mobile	Step-by-step guided questions replace chart lookup
No structured wound assessment prompts	STRIDE and TIMER frameworks built into every visit
No record of wound progression over time	Patient records stored on device; charts across visits
Multiple nurses, different devices	JSON export / import allows record sharing without a server
Dressing names only — no brand guidance	Brand name field built in; formulary can be added later

2 The Planning Sketch

How a hand-drawn sketch becomes an app

Before any code is written, a planning sketch maps out the user's journey — what they see first, what choices they make, and where they end up. The sketch below represents the kind of diagram that would be drawn on paper or a whiteboard in the early planning phase. Each box is a screen; each arrow is a navigation step; each diamond is a decision point.



3 The Three Frameworks

Clinical knowledge that underpins the tool

The tool is built on three established clinical frameworks. Understanding these is essential to understanding why each screen exists and what it contributes to the final recommendation.

Framework	What it stands for	What it tells us	Where it appears
STRIDE	Site · Tissue · Redness Inflection · Discharge · Edge	Structured description of the wound as it is today	Screens 2–7 (6 assessment steps)
TIMER	Tissue · Infection · Moisture Edge · Repair	Identifies barriers that are slowing or stopping healing	Screens 9–13 (5 assessment steps)
DoH Matrix	Dept of Health & Aged Care Consumable Choice Guide	Maps wound type + exudate + infection to dressing options	Results screen (26 clinical pathways)

4 Screen-by-Screen Flow

Every screen in the app and its purpose

#	Screen name	Phase	Key inputs / outputs	Feeds into
0	Welcome	—	Entry point; recent patient shortcut	Patient details or patient list
P	Patient details	Patient	UR number, name, DOB, clinician	STRIDE step 1
S1	Site & location	STRIDE	Body region selection	S2
S2	Tissue type	STRIDE	Necrotic / Slough / Granulating / Epithelial	S3 — also keys DoH lookup
S3	Surrounding skin	STRIDE	Multi-select peri-wound features	S4 — eczema triggers zinc note
S4	Infection	STRIDE	Yes / No + odour sub-question	S5 — keys DoH lookup
S5	Exudate level	STRIDE	Low / Moderate / High	S6 — keys DoH lookup
S6	Edge	STRIDE	Attached / Undermined / Rolled etc.	Dimensions screen
D	Dimensions	Measure	L x W x D in mm; auto surface area (mm sq)	TIMER T1 — stored for charts
T1	Tissue (TIMER)	TIMER	Non-viable tissue present? Yes / No	T2 — debridement flag
T2	Infection (TIMER)	TIMER	None / Local / Spreading / Systemic	T3 — escalation flags
T3	Moisture	TIMER	Dry / Balanced / Excess	T4 — absorbency modifier
T4	Edge advance	TIMER	Advancing / Stalled / Increasing	T5 — specialist flag
T5	Repair	TIMER	Progressing / Slow / Stalled	Results — referral flag
R	Results	Output	DoH pathway, primary + secondary dressings, clinical flags, vs-previous comparison	Patient history
H	Patient history	Records	All visits, 3 charts, accordion detail	New visit or export
PL	Patient list	Records	All patients, sorted by last visit	Patient history or new assessment

5 How the DoH Matrix Becomes a

From a printed chart to a lookup table

The DoH Consumable Choice Guide is a printed decision tree with 26 endpoints. In the planning phase, the key insight was that every endpoint can be uniquely identified by just three inputs:

The three-key lookup

KEY 1: Tissue type (5 options)

KEY 2: Exudate level (3 options: Low / Moderate / High)

KEY 3: Infection (2 options: Yes / No)

Combined, these produce up to 30 unique combinations — 26 of which correspond to real DoH pathways (e.g. slough + moderate + infected = Pathway 15).

This means the entire 26-pathway matrix can be stored as a simple lookup dictionary in code: given the three keys, instantly return the correct primary and secondary dressing options.

Example lookup key construction:

tissue type + '_' + exudate level + '_' + infected (yes/no)

Tissue	Exudate	Infected	Key produced	DoH Pathway
Slough	Moderate	Yes	slough_moderate_yes	15
Granulating	Low	No	granulating_low_no	20
Necrotic (ischemic)	High	Yes	necrotic-ischemic_high_yes	5
Epithelialising	Low	No	epithelialising_low_no	25

Additional output modifiers (applied after lookup):

Odour present	Append charcoal / carbon dressing note to recommendation
Venous eczema noted	Append zinc paste bandage availability note
Peri-wound maceration	Append moisture barrier / higher absorbency reminder
TIMER — spreading infection	Show red escalation alert on results screen
TIMER — non-viable tissue	Show red debridement alert on results screen
TIMER — stalled repair	Show blue specialist referral flag on results screen

6 Data Structure — What Gets Stored

How patient data is organised

Each patient has a record identified by their UR number. Each visit to that patient creates a new entry in their visit history. No data is sent to any server — everything lives on the nurse's device, in the browser's local storage.

Patient record (one per UR number)

UR number	Unique identifier — the record key
Patient name	Display name (optional)
Date of birth	For identification (optional)
Visits	Array of visit records (see right)

Each visit record contains:

date	ISO timestamp of assessment
clinician	Assessing nurse name
tissue	Tissue type selected
exudate	Low / Moderate / High
infected	Yes / No
dimL, dimW, dimD	Measurements in mm
area	Calculated surface area (mm sq)
skin	List of peri-wound features
odour	Yes / No
TIMER fields	tissue / infection / moisture / edge / repair
pathway	DoH pathway number (1–26)
primaryDressings	List of recommended primary options
notes	Free-text clinical notes

7 Longitudinal Tracking — How P

Seeing the wound journey over time

Once a patient has two or more visits recorded, the tool can compare assessments and show whether the wound is improving, static, or deteriorating. This involves two features: visit-to-visit comparison on the results screen, and charts on the patient history screen.

Visit-to-visit comparison (results screen)

Metric	How improvement is detected	Indicator shown
Surface area	Current area < previous area	Green arrow "23% smaller"
Tissue type	Progression toward epithelialising (Necrotic → Slough → Granulating → Epithelial)	Green "Progressing" or red "Deteriorating"
Exudate level	Current level lower than previous	Green "Reduced" or red "Increased"
Infection status	Changed from Yes to No	Green "Resolved" or red "Developed"

Longitudinal charts (patient history screen)

Chart	What it shows	Requires
Surface area trend line	Line graph of wound surface area (mm sq) across all visits with data points highlighted with dimensions recorded	2+ visits
Exudate level bar chart	Colour-coded bars (green=Low, amber=Moderate, red=High) for each visit	2+ visits
Tissue type timeline	Colour-coded blocks per visit; darker = less healed, teal = epithelialising	2+ visits

8 Technical Architecture in Plain

What kind of thing is this, technically?

Component	Plain English description	Technical term
Single file	The entire app is one file you can email or upload. No installation required.	Self-contained HTML/CSS/JS
No internet needed	Once the page is open, everything works offline. No server calls are made.	Client-side only application
Device storage	Patient records are saved in the browser. They persist between sessions.	Browser localStorage API

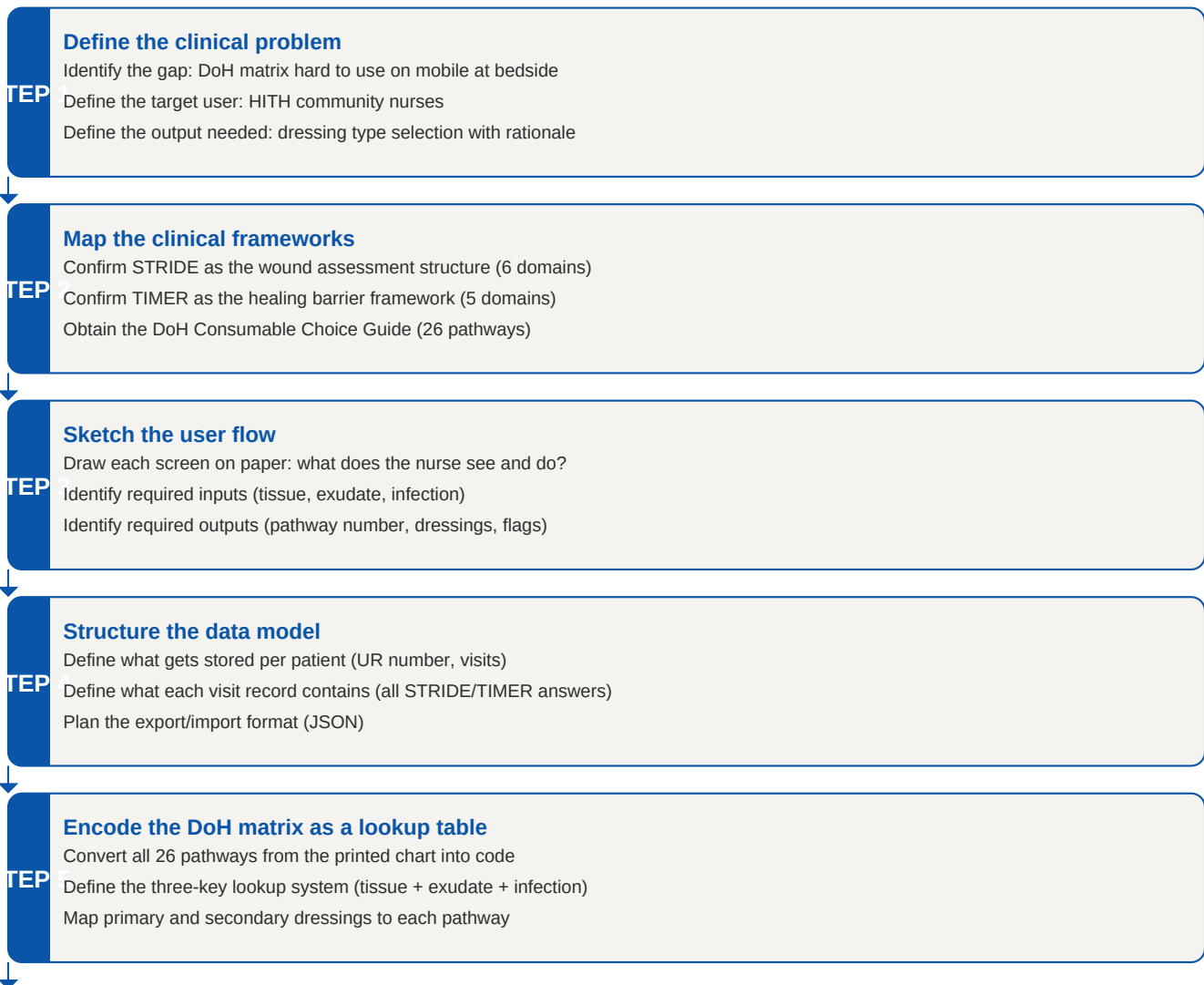
Export / import	Records can be saved to a file and loaded on a different device.	JSON serialisation / deserialisation
Charts	Graphs are drawn as scalable vector shapes, not images. They resize perfectly on any screen.	Inline SVG rendering
SharePoint hosting	The file sits in a document library like any Word or PDF document. Open the link — it runs.	Static file hosting
Brand colours	All colours are defined in one place at the top of the file. Change one line to retheme the tool.	CSS custom properties (variables)

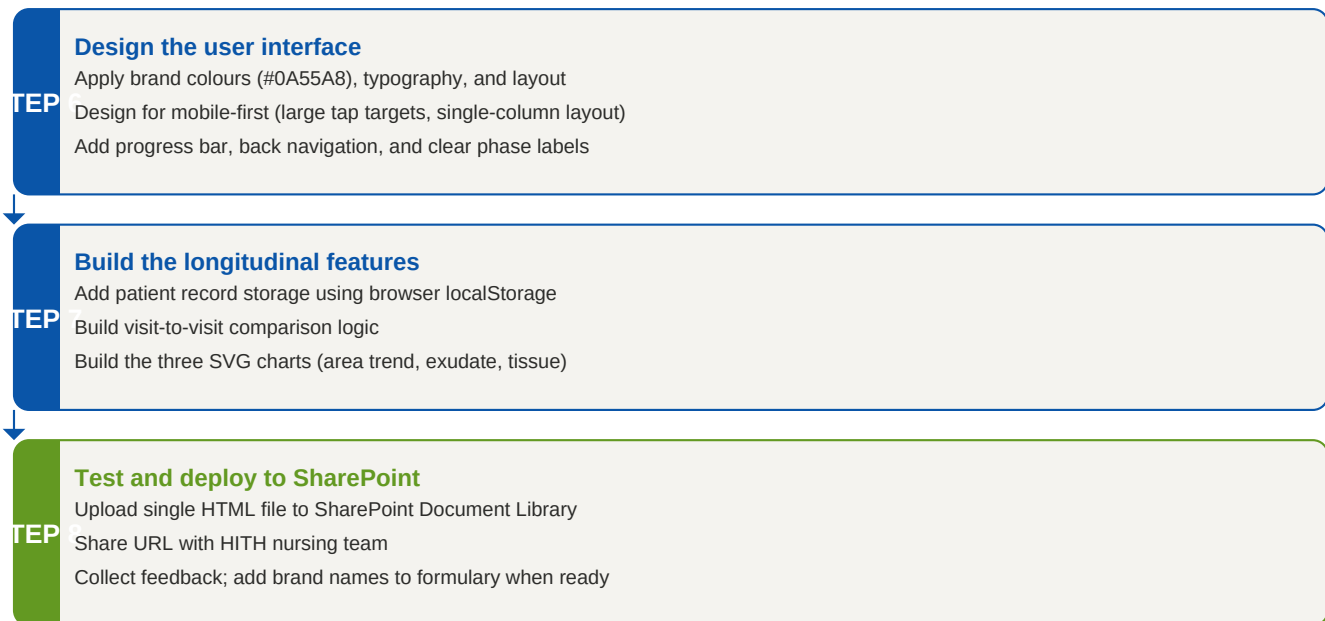
What the tool does NOT do

- x It does not send any data to the internet or to Anthropic.
- x It does not access a central database or patient management system.
- x It does not require a login, account, or internet connection to operate.
- x It does not replace clinical judgement — it is advisory only.
- x It does not store data across devices automatically (use export/import).

9 From Sketch to Deployed File

The planning-to-deployment pipeline





10 Glossary of Terms

STRIDE

Site · Tissue · Redness · Infection · Discharge · Edge. A structured wound assessment framework used to describe a wound systematically.

TIMER

Tissue · Infection/inflammation · Moisture · Edge · Repair/regeneration. A framework for identifying what is preventing a wound from healing.

DoH Matrix

The Australian Government Department of Health and Aged Care Wound Consumable Choice Guide — a 26-pathway decision tree mapping wound characteristics to appropriate dressing types.

Pathway

One of 26 numbered endpoints in the DoH matrix, each representing a unique combination of tissue type, exudate level, and infection status.

Exudate

The fluid (serum, pus, or blood) produced by a wound. Levels are classified as Low, Moderate, or High.

Granulating

A wound tissue type characterised by beefy-red, cobblestone-textured new tissue — a sign of healthy healing.

Epithelialising

The final stage of wound healing, where new pink skin forms at the wound edges or across the wound surface.

Slough

Yellow, grey, or tan devitalised (dead) tissue that can impede healing if not removed.

Necrotic

Dead tissue; may be ischemic (caused by inadequate blood supply) or non-ischemic.

LocalStorage

A feature built into all modern web browsers that allows a web page to save data on the device. Data persists after the browser is closed.

JSON

JavaScript Object Notation — a plain-text format for storing and transferring structured data. Used for the export/import feature.

SVG

Scalable Vector Graphics — a format for drawing shapes and charts using mathematical descriptions rather than pixels. Scales perfectly on any screen size.

Single-file app

An application where all code, styling, and logic is contained within one file. Nothing else needs to be installed or downloaded.

SharePoint

Microsoft's enterprise document management and collaboration platform, used here as a simple file hosting service for the HTML application.

UR number

Unit Record number — a unique identifier assigned to each patient within a health service.

Primary dressing

The dressing placed directly in contact with the wound bed, chosen to manage the wound's specific characteristics.

Secondary dressing

An outer dressing applied over the primary dressing to provide additional absorbency, protection, or security.

This document is an internal planning reference for the Wound Care Advisor tool. The tool itself is advisory only and must be used in conjunction with clinical judgement. All recommendations are based on the Australian Government Department of Health and Aged Care Wound Consumable Choice Guide.